

Chemistry Ch 5.3: Types of Bonds

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5.3.1 Ionic Bond

Ek atom electron **de** deta hai, doosra **le** leta hai. Dene wala **cation** (+) banta hai aur lene wala **anion** (–). Phir ye ulte charge **ek doosre ko kheenchte** hain, aur wohi ionic bond hai.

Kitab ki misalein: NaCl, MgF₂, MgO, NaF. Dikhane ke liye **electron dot aur cross** diagram banate hain.

5.3.2 Covalent Bond

Dono atoms electron **share** karte hain, koi deta leta nahin.

Single bond: ek jora share (H₂, F₂, CH₄, H₂O)

Double bond: do jore (O₂, CO₂) **Triple bond:** teen jore (N₂, HCN)

5.3.3 Polar aur Nonpolar, aur farq kaise pehchanein

Share karne ka matlab ye nahin ke **barabar** share ho raha hai.

Nonpolar: dono atoms ek jaise, electronegativity ka farq **nahin**, to electrons **beech** mein rehte hain.

Misal: H₂, O₂, N₂.

Polar: ek atom zyada electronegative, wo electrons **apni taraf** kheench leta hai. Us par halka **manfi** aur doosre par halka **musbat** charge aa jata hai. Misal: H₂O, HCl.

Paani ki saari khasoosiyat isi polarity se aati hain.

5.3.4 Coordinate Covalent Bond

Ye bhi share hi hai, magar ek farq ke saath: **dono electrons ek hi atom** dete hain. Doosra atom sirf **khaali jagah** deta hai.

Dene wale ke paas **lone pair** hona zaroori hai.

Kitab ki teen misalein: NH₄⁺ (ammonia nitrogen apna lone pair H⁺ ko deta hai), H₃O⁺ (paani ki oxygen), aur NH₃·BF₃.

Quiz

1. Explain how an **ionic** bond forms and how a **covalent** bond forms, with two examples of each from your book. Then name a molecule with a single, a double and a triple bond.
2. What is the difference between a **polar** and a **nonpolar** covalent bond? Then define a **coordinate covalent** bond and give the three examples your book provides.